



Lecture 11: Introduction to Power Plants

EEE 321: Power System I

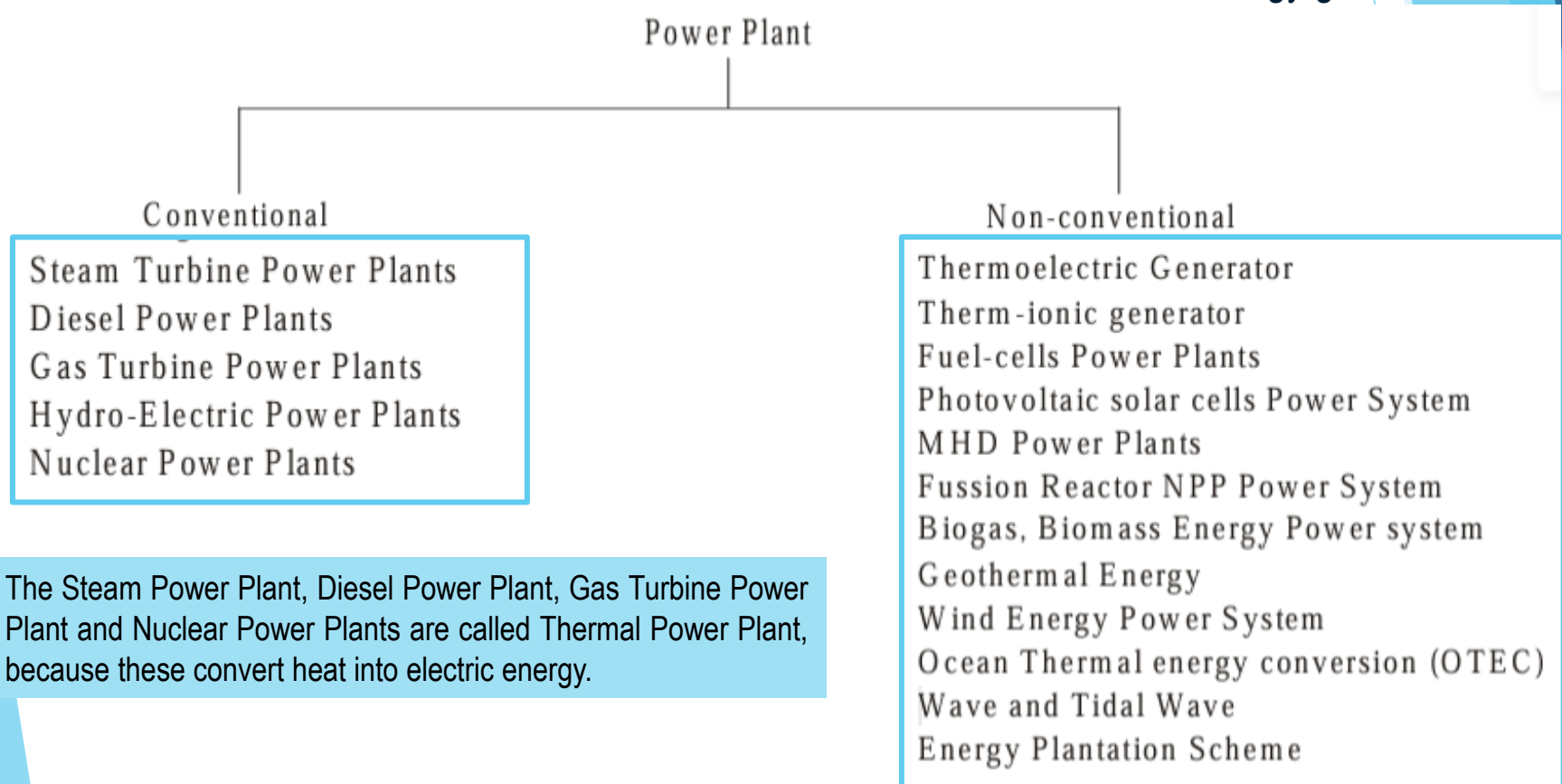
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Presentation Outline

- ❑ Introduction to Power Plants
- ❑ Various types of Power Plants- Advantages, Disadvantages and Operation
- ❑ Comparative Analysis
- ❑ Economics of Power Plant- Load Curves, Factors related to Load Curve
- ❑ Examples and Problems

Introduction to Power Plants

A power plant is assembly of systems or subsystems to generate electricity, i.e., power with economy and requirements. It may be defined as a machine or assembly of equipment that generates and delivers a flow of mechanical or electrical energy. The main equipment for the generation of electric power is generator. When coupling it to a prime mover runs the generator, the electricity is generated. The power plant itself must be useful economically and environmental friendly to the society. The present discussions are oriented to conventional as well as non-conventional energy generation.



Steam Turbine Power Plant

A steam turbine power plant generates electricity by converting heat energy into mechanical energy using a steam turbine, which then turns a generator. The process involves a boiler heating water to produce high-pressure steam, which expands through the turbine to spin its blades, creating mechanical power. This mechanical energy drives a generator, producing electricity.

Fuel Flexibility:

A variety of fuels, including coal, natural gas, oil, and biomass, can be used to generate steam.

Less Land Required:

Steam plants generally require less physical space than hydroelectric power plants.

Reliable Power Source:

They provide a consistent and dependable source of electricity, suitable for large-scale generation.

Advantages

High Running Costs:

Fuel costs and ongoing maintenance contribute to high operational and running expenses.

Low Efficiency:

Steam power plants typically have a low overall efficiency, around 35%-41%.

Air Pollution:

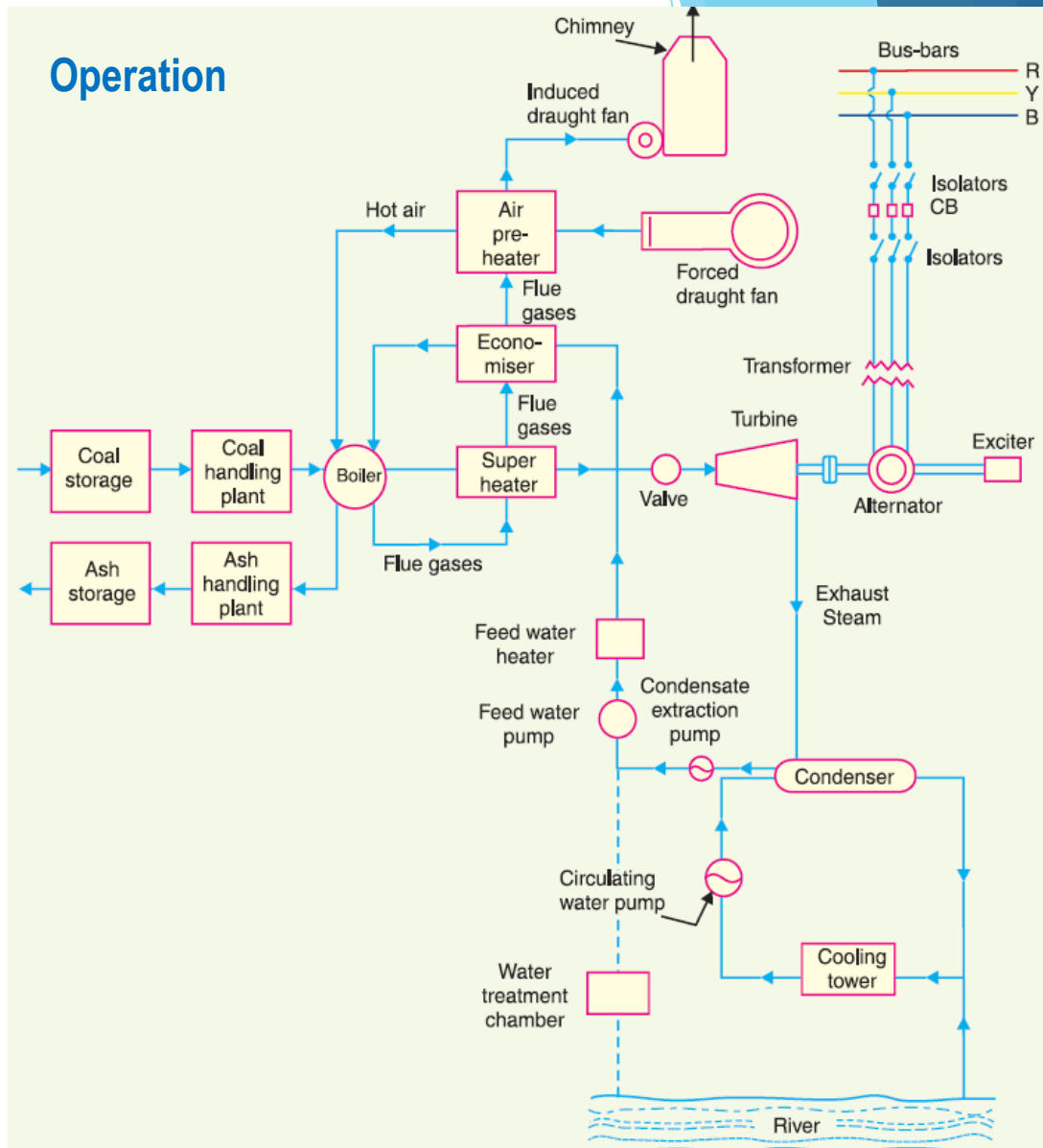
Burning fossil fuels releases harmful gases and fumes into the atmosphere.

Large Water Requirements:

A substantial amount of water is necessary for producing steam and for cooling the condenser.

Disadvantages

Operation



Diesel Power Plant

A diesel power plant converts chemical energy from diesel fuel into electrical energy using a diesel engine, which drives an electric generator. These plants are often used as main sources for small power needs, for backup or emergency power at hospitals and industries, and for peak loads.

Quick Startup:

Diesel generators can be started and stopped quickly, making them ideal for standby or emergency power.

Compact and Mobile:

They are smaller than other power plants of similar output and can be easily transported.

High Thermal Efficiency:

Diesel plants offer better thermal efficiency, especially at partial loads.

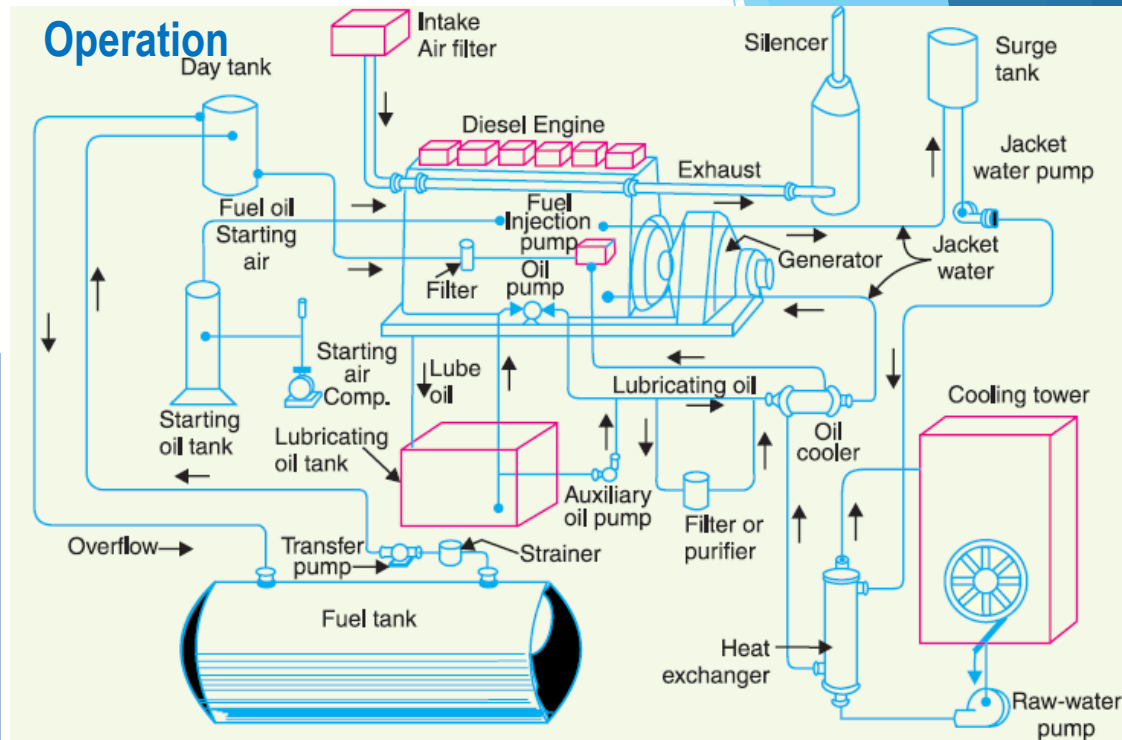
Low Capital Cost:

The initial investment required for a diesel power plant is lower compared to steam or hydro plants.

Low Water Requirement:

They need less water for cooling, which is beneficial in areas with limited water resources.

Advantages



High Running Costs:

Diesel fuel is expensive, leading to higher running costs for the plant.

High Emissions:

They produce significant greenhouse gas emissions and other pollutants.

High Maintenance and Lubrication Costs:

The costs associated with maintenance and the purchase of lubricants are generally high.

Limited Power Generation: Diesel plants typically generate small amounts of power, making them unsuitable for large-scale electricity needs.

Disadvantages

Gas Turbine Power Plant

A gas turbine power plant generates electricity by using natural gas to power a turbine, which in turn drives a generator. It consists of a compressor, combustion chamber, and turbine. While simple cycle turbines have 35-40% efficiency, combined cycle (CCGT) plants achieve up to 60% efficiency by using waste heat from the gas turbine to generate additional power. These plants are known for their quick start-up times, compact design, and are critical for both base-load and peak power supply.

Quick Starting & Loading:

They can start and reach full load capacity very rapidly, making them ideal for peak power demands.

Compact Size:

Gas turbines are smaller and lighter for the same power output compared to steam power stations.

Lower Capital Costs:

Initial costs can be lower than steam plants because they don't require a boiler.

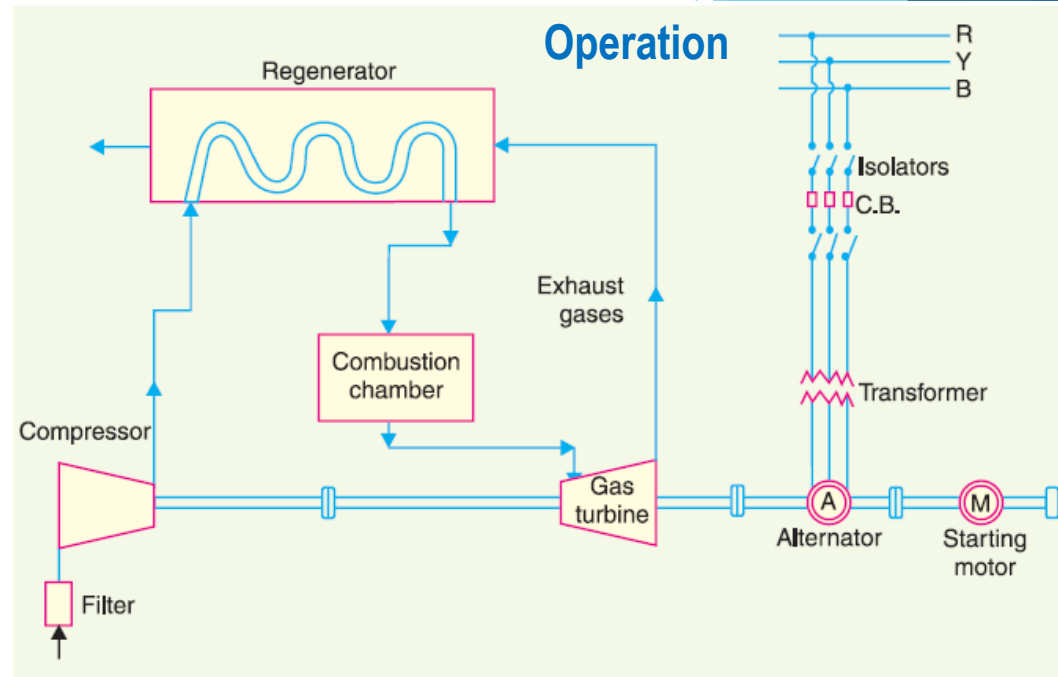
Reduced Water Usage:

They do not need a condenser, leading to less water consumption compared to steam plants.

Fuel Flexibility:

They can operate on a range of fuels, including natural gas and heavy oils

Advantages



Low Thermal Efficiency: Open-cycle gas turbines have low thermal efficiency, as a significant portion of the heat is lost in the exhaust. Efficiency can be improved in a combined cycle configuration.

High Operating Costs: Running costs can be high due to fuel type and price.

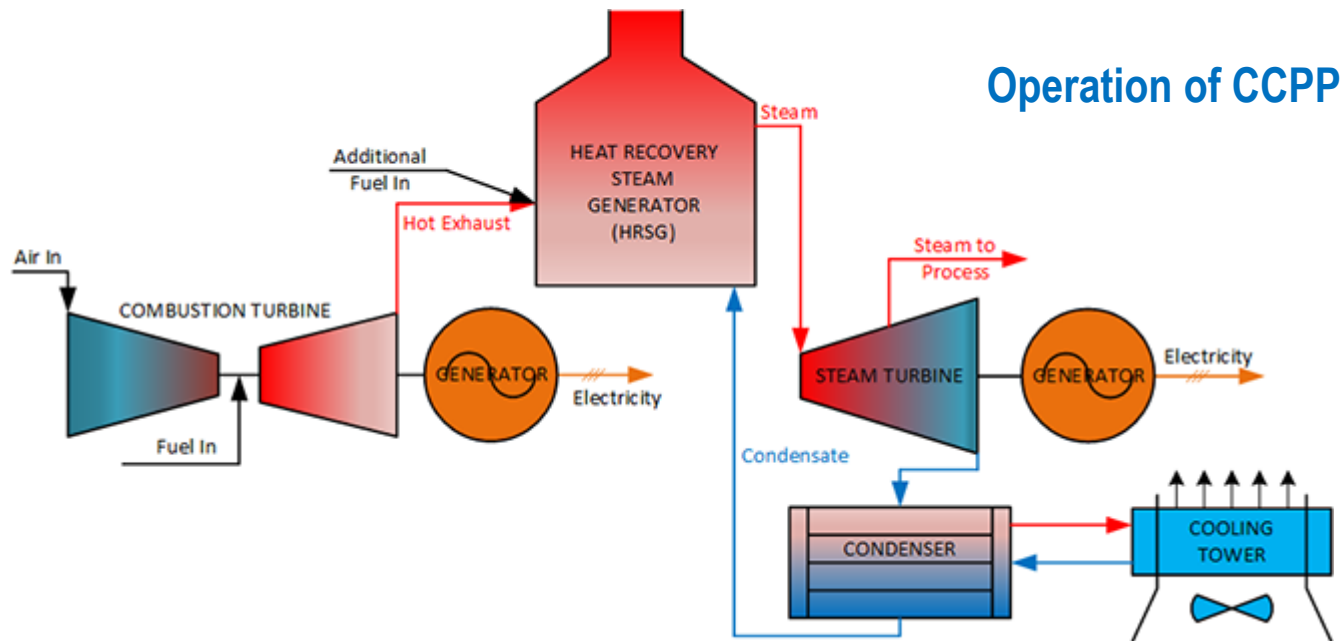
High Operating Temperatures: The extremely high temperatures required for operation necessitate expensive, special metal alloys and advanced cooling for turbine blades.

Component Sensitivity: The high temperatures and pressures make the turbine very sensitive to changes in operating conditions.

Short Component Life: The severe operating conditions of high temperatures and potential attack from combustion byproducts can lead to a shorter operational life for components.

Disadvantages

Gas Turbine Power Plant



A combined cycle power plant (CCPP) generates electricity by using both a gas turbine and a steam turbine in tandem, significantly increasing overall energy efficiency compared to simple cycle plants. It works by using a gas turbine to produce power, then capturing its hot exhaust gases in a heat recovery steam generator (HRSG) to create steam, which drives a steam turbine to generate additional electricity. This integrated process maximizes the use of fuel, leading to higher power output and reduced emissions.

Modern CCPPs achieve efficiencies exceeding 60%, with some reaching over 64%, significantly higher than conventional steam power plants which are around 40% efficient. This higher efficiency results in lower fuel costs and reduced greenhouse gas emissions compared to single-cycle plants.

Hydroelectric Power Plant

A hydro-electric power plant generates electricity using the flow of water, a process called hydroelectricity, by converting the kinetic energy of water into mechanical energy to spin turbines, which then drive a generator to produce electricity. These plants are a significant source of renewable energy, but their environmental and social impacts, high costs, and reliance on water availability are notable drawbacks.

Renewable Energy:

Relies on the natural water cycle, a sustainable and inexhaustible resource.

Low Emissions:

Produces virtually no greenhouse gases or air pollution once built, unlike fossil fuel plants.

Reliable and Dispatchable:

Provides a stable, predictable power supply, and can be quickly ramped up or down to meet changing energy demands.

Cost-Effective Long-Term:

After the high initial investment, operating costs are low and the plants have a long lifespan.

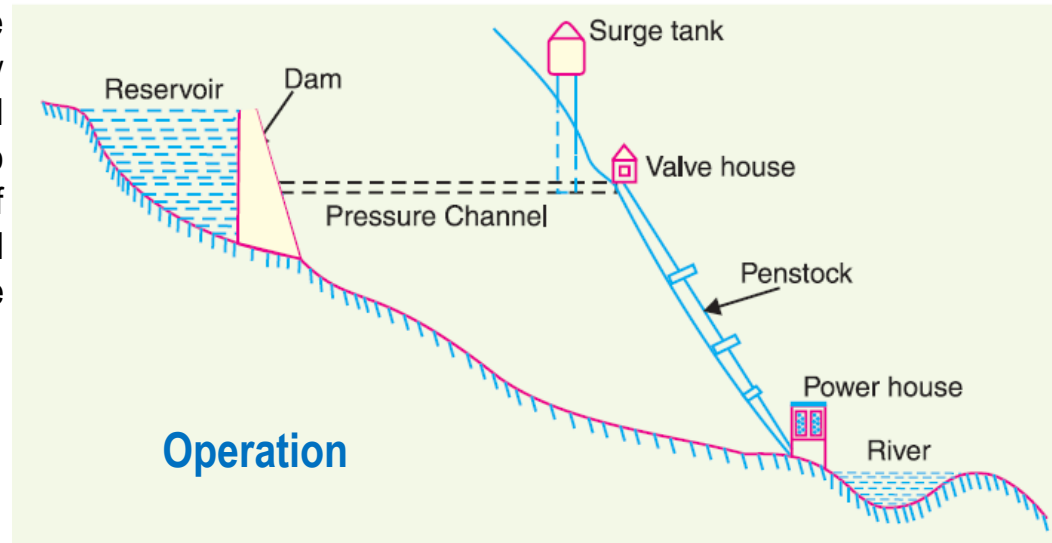
Water Storage:

Reservoir dams can store water for irrigation, drinking, and flood control.

Supports Other Renewables:

Provides operational flexibility and energy storage capabilities that complement other intermittent renewable sources like solar and

Advantages



High Initial Costs:

Building dams and power plants is extremely expensive and requires significant investment.

Limited Suitable Locations:

Only a limited number of sites worldwide are geologically suitable for dam and plant construction.

Environmental Impact:

Dams flood large areas, destroying ecosystems, displacing communities and wildlife, and altering river flow.

Disadvantages

Nuclear Power Plant

A nuclear power plant generates electricity by using heat from an ongoing atomic fission reaction in a reactor to boil water, which creates steam to spin a turbine connected to a generator. This process is similar to a conventional thermal power plant, but instead of burning fuel, it splits atoms of elements like uranium to release heat.

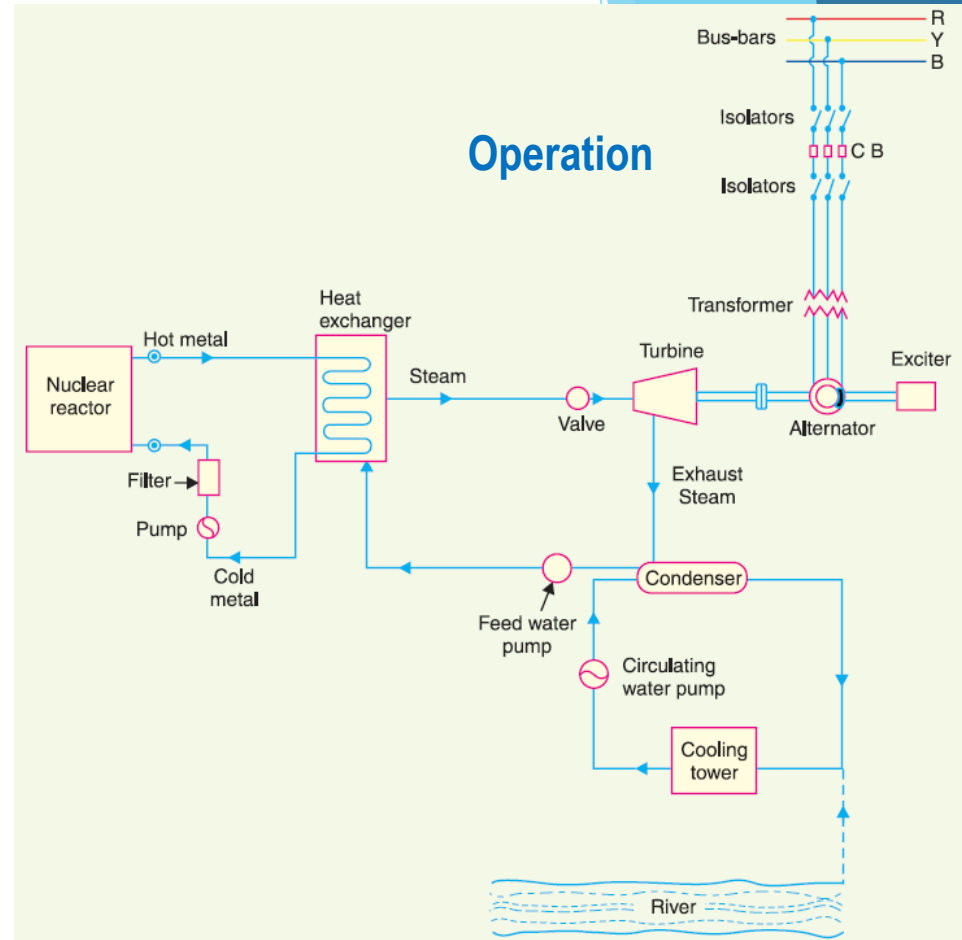
Low Greenhouse Gas Emissions: Nuclear power plants do not emit carbon dioxide, methane, or other greenhouse gases during operation. This makes them a key tool in combating climate change.

High Power Output & Efficiency: Nuclear plants produce a large amount of electricity from a small amount of fuel. A single uranium fuel pellet has as much energy as a ton of coal.

Reliable and Consistent Energy Source: Unlike intermittent renewable sources like wind and solar, nuclear plants can operate continuously, 24/7, providing a stable "baseload" power supply to the electrical grid.

Advantages

Small Land Footprint: Nuclear power plants require a relatively small amount of land compared to solar farms or wind farms for the same amount of electricity generation.



Radioactive Waste: Nuclear fission creates radioactive waste that remains hazardous for thousands of years. Safely storing and disposing of this waste is a major long-term challenge, as there are currently no permanent disposal sites in the U.S.

Disadvantages

High Upfront Costs: Building a nuclear power plant is extremely expensive, often costing billions of dollars. This can make them a less attractive investment compared to other energy sources.

Risk of Accidents: While highly regulated and rare, accidents like those at Chernobyl and Fukushima demonstrate the potential for catastrophic environmental and health consequences, including widespread radioactive contamination.

Security Concerns: The materials used in nuclear plants could potentially be diverted for the creation of nuclear weapons, and the plants themselves are potential targets for terrorist attacks.

Nuclear Power Plant

Feature	Steam Power Plant	Diesel Power Plant	Gas Turbine Plant	Hydroelectric Plant	Nuclear Plant
Principle	Water is heated by burning fossil fuels to create high-pressure steam, which spins a turbine to generate electricity.	A diesel engine acts as a prime mover, burning fuel to create mechanical energy that turns a generator.	Compressed air is mixed with natural gas, ignited, and the hot, high-pressure gas expands to spin a turbine.	The potential energy of stored water is converted into kinetic energy as it flows through a pipe, turning a turbine.	Heat from a controlled nuclear fission reaction boils water into steam, which then drives a turbine.
Fuel Source	Coal, petroleum, natural gas, biomass	Diesel, furnace oil	Natural gas, liquid fuels	Water (hydrological cycle)	Uranium-235, Plutonium-239, Thorium-232
Initial Cost	Moderate-High. The cost depends on the size and technology, but it's generally lower than hydro or nuclear.	Low. The initial setup is relatively cheap, making it suitable for temporary or backup power.	Low-Moderate. The cost is competitive and can be quickly installed.	Very High. Building a dam and reservoir requires a massive initial investment.	Very High. Requires significant investment in the nuclear reactor and safety infrastructure.
Running Cost	High. Consumable fuel (coal, oil) and extensive maintenance lead to high operating costs.	Highest. Diesel is expensive, making the running cost prohibitive for large-scale, continuous use.	Moderate-High. The cost is tied to the price of natural gas.	Very Low. No fuel cost and minimal operational expenses.	Very Low. A small amount of fuel produces a massive amount of energy.
Efficiency	~35-40%. The efficiency is limited by thermodynamic cycles and heat loss.	~30-35%. While generally more efficient than a steam plant, it's still a thermal system with heat loss.	~35-40%. Combined-cycle gas plants can achieve over 60% efficiency by using exhaust heat.	~85-90%. Highly efficient as there is very little energy loss.	~35-40%. Similar to steam plants, but with a much denser energy source.

Comparative Analysis

Feature	Steam Power Plant	Diesel Power Plant	Gas Turbine Plant	Hydroelectric Plant	Nuclear Plant
Starting Time	Very slow. Can take several hours to heat the boiler and get the plant running.	Very fast. Can be started and loaded almost instantly, making it ideal for peak loads or emergencies.	Fast. Can be brought online quickly, often in just a few minutes.	Fast. Can be started and shut down rapidly to meet sudden changes in demand.	Very slow. Starting and shutting down a nuclear reactor is a lengthy, complex process.
Space Required	Large. Requires space for the boiler, turbines, and fuel storage (e.g., coal yards).	Small. Highly compact, requiring minimal space.	Small. Compact and can be built quickly.	Very Large. Requires a large area for the dam and the creation of a reservoir.	Large. Needs significant space for the reactor, containment building, and safety systems.
Pollution	High. Produces significant air pollutants (CO ₂ , SO _x , NO _x) and thermal pollution to water bodies.	High. Emits pollutants, soot, and noise.	Moderate. Produces fewer emissions than coal/diesel but still releases CO ₂ and NO _x .	None (Air). Does not produce any greenhouse gas or air pollutants during operation.	None (Air). Releases no greenhouse gases or air pollutants during operation.
Disadvantages	High pollution, slow to start, and high running costs.	Very high fuel cost, high emissions, and limited to smaller capacities.	High running cost, produces greenhouse gases, and is dependent on natural gas prices.	Dependent on water availability, has major environmental impacts, and is location-specific.	Produces radioactive waste, has high safety risks, and has a very high initial cost.
Applications	Base load. Provides a consistent and reliable supply of power to the grid.	Peak load/Backup. Used for emergency power, small-scale industrial use, and remote locations.	Peak load. Excellent for meeting sudden surges in demand.	Base & Peak load. Can provide both a consistent supply and rapid response to demand.	Base load. Provides a large, reliable source of continuous power for the grid.

Economics of Power Plant

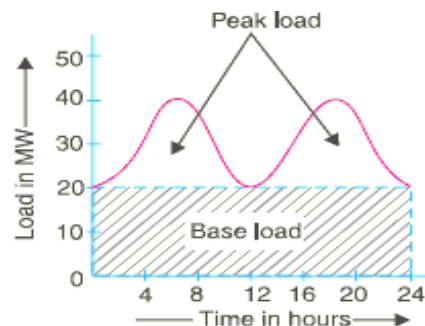
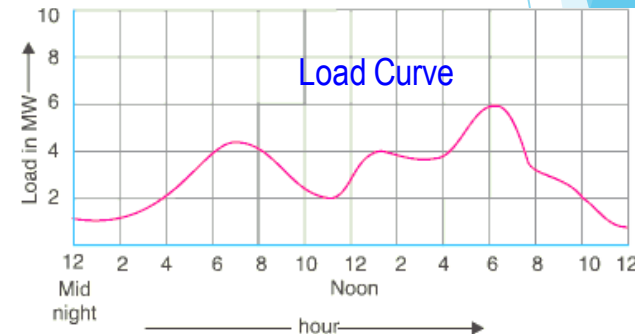
Variable Load on Power Station: Load Curve

- Power demands of different consumers vary in accordance with their activities.
- The result of this variation in demand is that load connected on a power station is never constant, rather it varies from time to time.

The load connected on a power station varies from time to time due to uncertain demands of the consumers and is known as **variable load on the station**.

The curve showing the variation of load on the power station with respect to (w.r.t) time is known as a **load curve**.

- Daily load curve
- Monthly load curve
- Yearly load curve
- Different types loads-
 - Residential load
 - Commercial load
 - Industrial load
 - Office Building load
 - Agricultural load
 - Institutional load
 - Municipal load



Base load. The unvarying load which occurs almost the whole day on the station is known as **base load**.

Peak load. The various peak demands of load over and above the base load of the station is known as **peak load**.

Economics of Power Plant

Importance of Load Curve

(i) The daily load curve shows the variations of load on the power station during different hours of the day.

(ii) The area under the daily load curve gives the number of units generated in the day.

Units generated/day = Area (in kWh) under daily load curve.

(iii) The highest point on the daily load curve represents the maximum demand on the station on that day.

(iv) The area under the daily load curve divided by the total number of hours gives the average load on the station in the day.

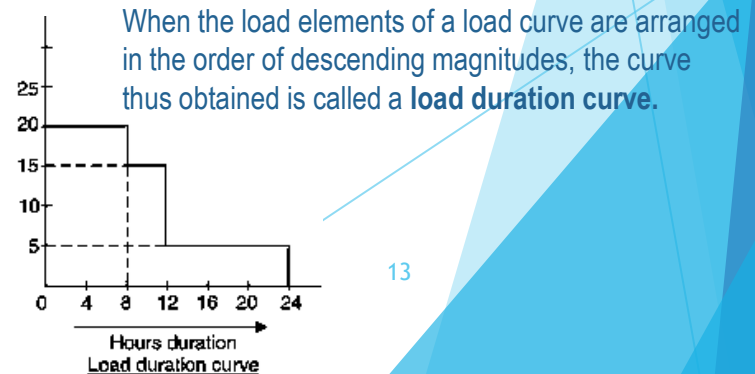
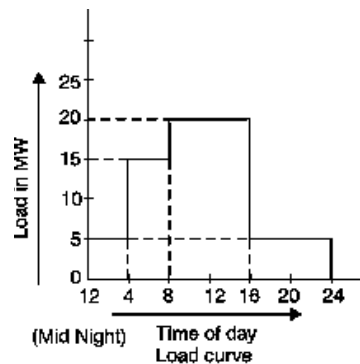
Average load = Area (in kWh) under daily load curve / 24 hours

(v) The ratio of the area under the load curve to the total area of rectangle in which it is contained gives the load factor.

Load factor = Average load / Max. demand

(vi) The load curve helps in selecting the size and number of generating units.

(vii) The load curve helps in preparing the operation schedule of the station.



Economics of Power Plant

Important factors from Load Curve

Connected load. It is the sum of continuous ratings of all the equipment connected to supply system.

For instance, if a consumer has connections of five 100-watt lamps and a microwave of 1500 watts, then connected load of the consumer is $5 \times 100 + 1500 = 2000$ watts.

Maximum demand: It is the greatest demand of load on the power station during a given period.

Demand factor: It is the ratio of maximum demand on the power station to its connected load. The value of demand factor is usually less than 1. **Demand factor = Maximum demand/Connected load**

Average load: The average of loads occurring on the power station in a given period (day or month or year) is known as average load or average demand.

Daily average load = No. of units (kWh) generated in a day/24 hour

Monthly average load = No. of units (kWh) generated in a month/Number of hours in a month

Yearly average load = No. of units (kWh) generated in a year/8760 hours

Load factor: The ratio of average load to the maximum demand during a given period is known as **load factor**.

Load factor = Average load/Max. demand

Higher the load factor of the power station, lesser will be the cost per unit generated. It is because higher load factor means lesser maximum demand. The station capacity is so selected that it must meet the maximum demand. Now, lower maximum demand means lower capacity of the plant which, therefore, reduces the cost of the plant.

Diversity Factor: The ratio of the sum of individual maximum demands to the maximum demand on power station is known as diversity factor i.e., **Diversity factor = Sum of individual max. demands/Max. demand on power station**

A power station supplies load to various types of consumers whose maximum demands generally do not occur at the same time.

Plant Capacity Factor: It is the ratio of actual energy produced to the maximum possible energy that could have been produced during a given period. i.e., **Plant capacity factor = Actual energy produced/Max. energy that could have been produced.**

Plant use factor: It is defined as the ratio of energy produced in a given time to the maximum possible energy that could have been produced during the actual number of hours the plant was in operation. Expressing the definition mathematically, **Plant use factor = $E/C t'$** where t' = Actual number of hours the plant has been in operation.

Economics of Power Plant

Examples

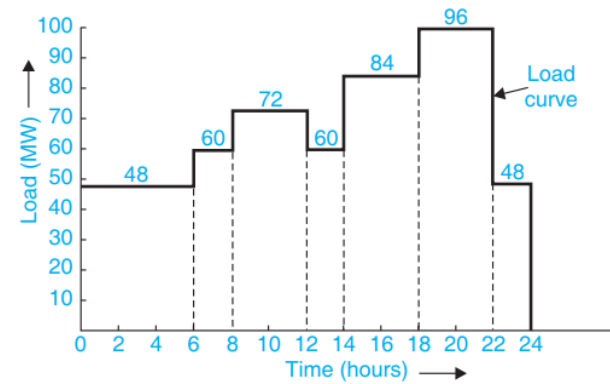


Fig. 9.13

Example 9.1. The maximum demand of a power station is 96000 kW and daily load curve is described as follows :

Time hours	0–6	6–8	8–12	12–14	14–18	18–22	22–24
Load (MW)	48	60	72	60	84	96	48

(i) Determine the load factor of power station.

(ii) What is the load factor of standby equipment rated at 30 MW that takes up all load in excess of 72 MW ?

Solution. Load curve is shown in Fig. 9.13.

$$\begin{aligned}
 \text{Energy generated} &= \text{area under the load curve} \\
 &= 48 \times 6 + 60 \times 2 + 72 \times 4 + 60 \times 2 \\
 &\quad + 84 \times 4 + 96 \times 4 + 48 \times 2 \\
 &= 1632 \text{ MWh} = 1632 \times 10^3 \text{ kWh.}
 \end{aligned}$$

(i) **Load factor :**

$$\text{Average load} = \frac{1632 \times 10^3}{24} = 68000 \text{ kW}$$

$$\text{Maximum demand} = 96000 \text{ kW} \quad (\text{given})$$

$$\begin{aligned}
 \therefore \text{Load factor} &= \frac{\text{Average load}}{\text{Maximum demand}} = \frac{68000}{96000} \\
 &= \mathbf{0.71. \quad (\text{Ans.})}
 \end{aligned}$$

(ii) **Load factor of standby equipment :**

The standby equipment supplies

$$84 - 72 = 12 \text{ MW for 4 hours (14 – 18)}$$

$$96 - 72 = 24 \text{ MW for 4 hours (18 – 22)}$$

$$\begin{aligned}
 \therefore \text{Energy generated by standby equipment} \\
 &= (12 \times 4 + 24 \times 4) \times 10^3 \\
 &= 144 \times 10^3 \text{ kWh}
 \end{aligned}$$

Time for which standby equipment remains in operation (from the load curve)

$$= 4 + 4 = 8 \text{ hours}$$

$$\text{Average} = \frac{144 \times 10^3}{8} = 18 \times 10^3 \text{ kW}$$

$$\text{Load factor} = \frac{18 \times 10^3}{24 \times 10^3} = \mathbf{0.75. \quad (\text{Ans.})}$$

Economics of Power Plant

Examples

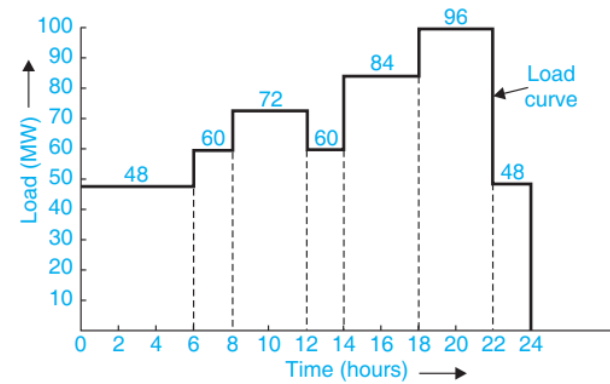


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$$\text{Load factor} = \frac{18 \times 10^3}{24 \times 10^3} = \mathbf{0.75. \quad (Ans.)}$$

Economics of Power Plant

Examples

Example 9.4. A generating station has a maximum demand of 5000 kW, and the daily load on the station is as follows :

Load (MW)	1000	1750	4000	1500
Time	11 PM to 8 AM	6 AM to 8 AM	8 AM to 12.00 Noon	12 PM to 1 PM
Load (MW)	3750	4250	5000	2250
Time (hours)	1 PM to 5 PM	5 PM to 7 PM	7 PM to 9 PM	9 PM to 11 PM

- Draw the load curve.
- Draw the load duration curve.
- Select the size and number of generator units.
- What reserve plant would be necessary ?
- Load factor.
- Plant capacity factor.

Solution. (i) **Load curve** is shown in Fig. 9.16 (a).

(ii) **Load duration curve** is shown in Fig. 9.16 (b).

(iii) **Size and number of generator units :**

From the load duration curve it is evident that generating sets of capacity 1000 kW, 1500 kW and 2500 kW will fulfil the requirement.

(iv) **Reserve capacity :**

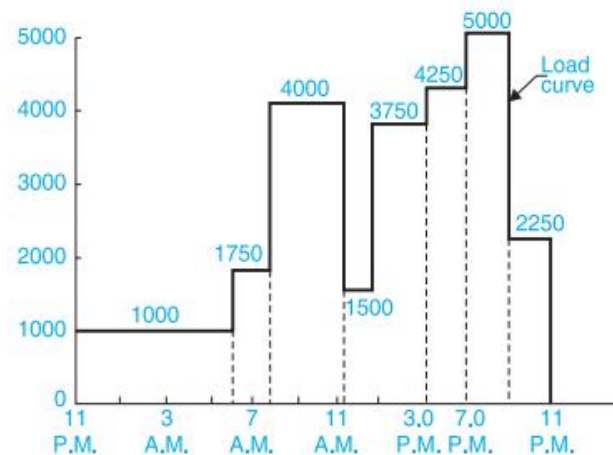
Also, reserve capacity = largest size of the unit in the station = **2500 kW.** (Ans.)

(v) **Load factor :**

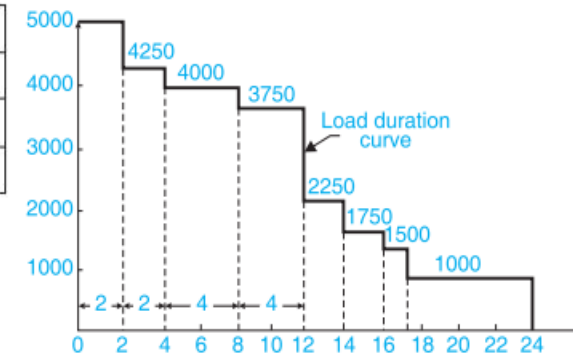
Area under the load curve gives the energy generated during 24 hours

$$\begin{aligned}
 &= 1000 \times 7 + 1750 \times 2 + 4000 \times 4 + 1500 \times 1 \\
 &\quad + 3750 \times 4 + 4250 \times 2 + 5000 \times 2 + 2250 \times 2 \\
 &= 7000 + 3500 + 16000 + 1500 + 15000 + 8500 \\
 &= 66000 \text{ kWh}
 \end{aligned}$$

$$\begin{aligned}
 \text{or Average load} &= \frac{66000}{24} = 2750 \text{ kW} \\
 \therefore \text{Load factor} &= \frac{\text{Average load}}{\text{Maximum demand}} = \frac{2750}{5000} \\
 &= 0.5. \text{ (Ans.)}
 \end{aligned}$$



(a) Load curve



(b) Load duration curve

Fig. 9.16

(vi) **Plant capacity factor :**

Plant capacity factor

$$\begin{aligned}
 &= \frac{E}{C \times t} = \frac{66000}{(2500 + 1500 + 1000 + 2500) \times 24} \\
 &= 0.367. \text{ (Ans.)}
 \end{aligned}$$

Economics of Power Plant

Examples

Example 9.5. A 60 MW power station has an annual peak load of 50 MW. The power station supplies loads having maximum demands of 20 MW, 17 MW, 10 MW and 9 MW. The annual load factor is 0.45. Find :

- (i) Average load.
- (ii) Energy supplied per year.
- (iii) Diversity factor.
- (iv) Demand factor.

Solution. Capacity of power station = 60 MW

Maximum demand on power station = 50 MW

(i) **Average load :**

$$\text{Load factor} = \frac{\text{Average load}}{\text{Maximum demand}}$$

i.e., $0.45 = \frac{\text{Average load}}{50}$

$$\therefore \text{Average load} = 50 \times 0.45 = \mathbf{22.5 \text{ MW. (Ans.)}}$$

(ii) **Energy supplied per year :**

Energy supplied per year

$$\begin{aligned} &= \text{Average load} \times \text{number of hours in one year} \\ &= (22.5 \times 10^3) \times 365 \times 24 \\ &= \mathbf{197.1 \times 10^6 \text{ kWh. (Ans.)}} \end{aligned}$$

(iii) **Diversity factor :**

Diversity factor

$$\begin{aligned} &= \frac{\text{Sum of individuals maximum demands}}{\text{Simultaneous maximum demand}} \\ &= \frac{20 + 17 + 10 + 9}{50} = \frac{56}{50} = 1.12 \end{aligned}$$

Hence diversity factor = **1.12. (Ans.)**

(iv) **Demand factor :**

$$\begin{aligned} \text{Demand factor} &= \frac{\text{Maximum demand}}{\text{Connected load}} \\ &= \frac{50}{20 + 17 + 10 + 9} = \frac{50}{56} = 0.89 \end{aligned}$$

Hence, demand factor = **0.89. (Ans.)**

Economics of Power Plant

Examples

Example 9.9. The following data relates to a steam power plant :

Maximum demand	= 30000 kW
Load factor	= 0.42
Coal consumption	= 1.1 kg/kWh
Boiler efficiency	= 84%
Turbine efficiency	= 88%
Price of coal	= \$70 per tonne

Determine the following :

- (i) Thermal efficiency of the plant.
- (ii) Coal bill of the plant for one year.

Solution. (i) Thermal efficiency of the plant :

$$\begin{aligned}\text{Thermal efficiency of the plant} &= \text{Boiler efficiency} \times \text{turbine efficiency} \\ &= 0.84 \times 0.88 = \mathbf{0.7392 \text{ or } 73.92\%}. \quad (\text{Ans.})\end{aligned}$$

(ii) Coal bill :

$$\begin{aligned}\text{Average demand on station} &= \text{Maximum demand} \times \\ \text{load factor}\end{aligned}$$

$$= 30000 \times 0.42 = 12600 \text{ kW}$$

$$\begin{aligned}\therefore \text{Energy generated per year} &= 12600 \times (365 \times 24) \text{ kWh}\end{aligned}$$

$$\begin{aligned}\therefore \text{Coal consumption} &= 12600 \times (365 \times 24) \times 1.1 \text{ kg per year}\end{aligned}$$

$$\begin{aligned}\therefore \text{Coal bill} &= \frac{12600 \times (365 \times 24) \times 1.1 \times 70}{1000}\end{aligned}$$

$$= \$ \mathbf{8498952}. \quad (\text{Ans.})$$

Check solutions of the following examples from
A Textbook Of Power Plant Engineering, R.K.Rajput
Chapter 9- 9.1 to 9.9

References:

'Principles of Power System', V.K. Mehta and Rohit Mehta, Chapter 2.

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QUESTIONS AND DISCUSSIONS

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